Bocchino Daniele

Software Engineer | Cloud & Data Enthusiast

LinkedIn | portfolio | GitHub | Lugano, Switzerland | bocchino.daniele@gmail.com | +41 79 514 14 97

Skills

- Python | Typescript | Spark | Flutter | React | SQL | Git | Docker | Ruby | Terraform | Kubernetes
- AWS | Azure | Cloud Computing | Machine Learning | Data Engineering | Web Development | Mobile Development
- Data Engineer | Cloud Architecture | Frontend | Backend | Full-Stack | Italian, English

Experiences

Software Engineer — **UBS AG** — **Switzerland** — *Oct. 2024 - Present*

- Drive optimization and reliability of large-scale statistical and risk models for the US mortgage market, leading IT initiatives for the Mars Pod.
- Architect and enhance model pipelines using Python, Spark, and Polars, achieving a 33% reduction in key computation runtime.
- Design and maintain CI/CD pipelines in GitLab and Kubernetes-based environments, enabling automated, scalable, and fault-tolerant deployments.
- Build specialized engineering tools for QRM activities (e.g., KMPI, simulation utilities) to accelerate analysis and improve operational efficiency.
- Developed and delivered a distributed, event-driven worldwide platform using **React**, **Node.js**, and **Python**, spanning frontend, backend, and microservices with a clean-code, maintainable architecture.

Software Developer Specialist — **SGR Compliance** — **Lugano, Switzerland** — *Jun. 2023 - Sep. 2024*

- As a project Lead, I plan, manage, and create infrastructure using **Terraform**, **AWS** services, and development for a disruptive SGR compliance project, leveraging and a full stack technology suite that includes **React**, **Java**, and containerization using **Docker** and **Kubernetes**
- Design and implementation applications using disruptive technologies such as: **AWS Cloud Services** (**Lambda**, **ECS**, **ECR, Amplify, S3, RDS, DynamoDB, LoadBalancer, API Gateway…**) and **ML & AI** solutions used on mobile application such as: **Text Recognition**, **Image Detection**…
- Design and develop full stack solution, web and mobile, using **Flutter**, **React** and **Ruby on Rails**
- Integrate applications using **REST API**-led and **cloud-first** architectural design patterns

Internship Trainee - Master's Degree — **PhuseLab** — **Milan, Italy** — *Nov. 2022 - July 2023*

- Evaluation and selection of optimal algorithms to address the problem of **Contextual Multi-Armed Bandits** in the real-world context of audiovisual attention, with emphasis on **Thompson's sampling algorithm**
- Implemented a machine learning model in Python using ML and data analysis libraries such as : **Pandas**, **Numpy**, **Scikit-learn**, **Pytorch**

Freelancer Web & Mobile — **Italy** — *Jul. 2022 - Feb. 2023*

- Developed a mobile application CIUKA - Party Game using **Dart** on the framework **Flutter**.
- Created a web application Lora Giusta with over 10 thousand views in 2022 for a bed and breakfast near Como increasing l number of customers by over 70%. The Application was realized with **Javascript**, **ReactJs**, **Node**, and the css framework **Bulma.**

Research Collaborator - PRIN 2017 & VAST — **ISLab** — **Milan, Italy** — *Jun. 2021 - Mar. 2022*

- Won the University of Milan competition to create a Medilett web application for the MIUR PRIN 2017 project funded by the Ministry of Education, University and Research (2019-2022) with the aim of creating a PROSOPOGRAPHIC ATLAS OF MEDIEVAL ROMAN LITERATURE
- Won the University of Milan competition to create a ISLab web application for the European Project VALUES ACROSS SPACE AND TIME (VAST).
- The backend of the application was built with **Python**, with **Flask,** to define the REST API. The front end was built in **ReactJs** with the help of the css framework **Chakra UI.** All information has been saved on **PostgreSQL DB** and **MongoDB.**

Internship Trainee - Bachelor's Degree — **ISLab** — **Milan, Italy** — *Sept. 2020 - Feb. 2021*

- I built a web application in **ReactJs** capable of analyzing text documents and highlighting verbs, adjectives, pronouns, etc.. The Backend, built in **Python** with **Flask**, constituted the REST API of the application and all Natural Language Processing (**NLP**) operations.

Projects

- **Cloud Phoenix:** the "cloud-phoenix" project builds the Phoenix Application's infrastructure using **Terraform (IaC)** on **AWS**, incorporating AWS services such as: **ECS**, **ECR**, **DocumentDB**, **CloudWatch** and **CI/CD** using **CodeCommit**, **CodeBuild** and **CodePipeline** for deployment.
- **Cloud Gaming:** It's a project for the **Business Process Engineer** exam,it included analysis with **BPMN 2.0** and drafting an **AWS** architecture.

Education

Master's Degree of Computer Science — **UniMi** — **Milan, Italy** — Sept. 2021 - Jul. 2023

- *Advanced Biometrics | Affective Computing | Artificial Intelligence | Biometric Systems | Business Information Systems | Business Process Engineering | Intelligent Systems for Industry, Supply Chain and Environment | Multimedia Architectures | Natural Interaction | Privacy and Data Protection | Web Algorithmic*

Bachelor's Degree of Computer Science — **UniMi** — **Milan, Italy** — Sept. 2016 - Feb. 2021

- *Algorithms and Data Structures | Automata and Formal Languages | Computer Architecture I | Computer Architecture II | Computer Programming | Continuum Mathematics | Computer Networks | Cryptography I | Databases | Digital Marketing | Digital Products Law | Discrete Mathematics | Information Systems | Information & Transmission Theory | Mathematical Logic | Operating Systems | Security and Privacy | Software Engineering | Statistics and Data Analysis*

Certifications

- **Microsoft - Azure Data Fundamentals** — *Aug. 2024*
- **Microsoft - Azure AI Fundamentals** — *Aug. 2024*
- **Microsoft - Azure Fundamentals** — *Jul. 2024*
- **Hashicorp - Terraform Associate** — *Oct. 2023*
- **IBM Data Science Professional Certificate** — *Mar. 2023*
- **AWS Certified Solutions Architect - Associate** — *Feb. 2023*
- **AWS Certified Developer - Associate** — *Feb. 2023*
- **AWS Certified Cloud Practitioner** — *Dec. 2022*